Database Systems

Design Project

Bingo Book : Shinobi Intelligence Database

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Abstract

In the world of shinobi-warriors trained in the art of ninjutsu,taijutsu, and genjutsu, keeping intel on different shinobis, their abilities, their ranks, and similar attributes is crucial. Where the global order of the ninja world is at stake, large scale ninja wars, vicious ninja operations, and competitive tournaments, chunin exams are an all time reality. Nations need to have complete information on shinobis of other nations to estimate the strength of each nation. Such estimates help in battlefields,competitions, and missions to anticipate the scale of defense or offense required on a certain forefront. Moreover, it is also imperative to keep intel on rogue ninjas and associations-akatsuki to be able to tackle their villainous agendas. Lastly, tracking the progress skills of shinobis trained in the academy is essential for predicting future military capabilities of one's own nation. In a nutshell, a well-structured ninja database is the bread and butter of today's ninja world. An efficient database means a stronger and well informed nation, technically enhancing its capability to act strategically and have a tactical advantage over others.

Description

In the ninja world, all novice ninjas enroll in an academy and graduate after passing first tier exams known as genin exams. After which they are assigned in teams of three under supervision of senior ninjas to further train and enhance their ninja prowess. If deemed fit, the senior ninja trainer (jonin), approves his/her students to partake in second tier exams, known as chunin exams where genins from different nations come to compete for the title of chunin. This project aims to maintain shinobi database of :

1. Novice shinobis enrolled in academy of different nations
2. Shinobis that passed the Genin exams
3. Shinobis assigned to different rank missions
4. Shinobis Shinobis participating in Chunin exams.

Not all shinobis conform to the structures of the ninja world and its ideologies. Not all are courageous enough to walk the path of righteousness. So the fallen ones, the ones who betray their nations and resort to unlawful practices, need to be hunted down and put to rest for the good of all.

5. This project also aims to maintain a bingo book, keeping intel on any shinobis that have gone rogue to do the aforementioned deed.

Moreover, we have incorporated additional utilities to keep intel on resources of different nations, which includes ninja weapons, forbidden scrolls and other ninjutsu sources. Thus, this project also:

6. Maintains track of village’s military arsenals and ninjutsu resources.

Design Scope

Entities involved in the database are :

1. Shinobi
2. Village
3. Jutsu
4. Clan
5. Rogue Association
6. Ninja Mission
7. Genin Squad
8. Standoff
9. Chunin Competition
10. Inventory

Some other Entities have also been added which act as bridge Tables between any two of the above entities :

1. Presided_By : Links *Shinobi* and *Village*
2. Led_By : Links *Shinobi* and *Clan*
3. Created_By : Links *Shinobi* and *Jutsu*

Design's Schema

Shinobi:

1. **ninjaId** : integer, primary key to uniquely identify each ninja
2. ninjaName : varchar(50)
3. dateOfBirth : date
4. NinjaStatus : varchar(50), represents current status of the ninja (Alive, Deceased, Rogue etc.)
5. *clan* : varchar(50), foreign key referencing the clan the ninja belongs to
6. chakraNature : varchar(50)
7. *jutsuName* : varchar(50), foreign key referencing the signature jutsu of the ninja
8. *village* : varchar(50), foreign key referencing the village the ninja resides in
9. ninjaRank : varchar(50)
10. *mentorNinjaId* : integer, nullable foreign key self-referencing shinobi to indicate who mentored this ninja

Village:

1. **villageName** : varchar(50), primary key to uniquely identify each village
2. region : varchar(50), the land or territory the village belongs to
3. foundedYear : integer
4. VillagePopulation : integer
5. tailBeast : varchar(10), the tailed beast associated with the village if any
6. powerIndex : decimal(3,2), a numeric rating representing the overall strength of the village

Presided_By:

1. ***villageName*** : varchar(50), part of composite primary key, foreign key referencing village
2. ***ninjaId*** : integer, part of composite primary key, foreign key referencing the ninja who leads the village
3. leaderShipStartingDate : date, the date this ninja assumed leadership
4. LeaderShipStatus : varchar(50), indicates whether the leadership is Active or ended

Jutsu:

1. **jutsuName** : varchar(50), primary key to uniquely identify each jutsu
2. jutsuType : varchar(50), the category of jutsu (Ninjutsu, Taijutsu, Genjutsu etc.)
3. coolDownTime : decimal(5,2), time required before the jutsu can be used again
4. powerLevel : decimal(3,2), a numeric rating of how powerful the jutsu is

5. jutsuStatus : varchar(50), indicates whether the jutsu is Active, Obsolete or a Clan Technique
6. chakraConsumption : varchar(50), level of chakra required (Low, Medium, High, Extreme)
7. rankRequired : varchar(50), minimum ninja rank needed to perform this jutsu
8. elementalNature : varchar(50), the elemental affinity of the jutsu if any

Created_By:

1. ***NinjaID*** : integer, part of composite primary key, foreign key referencing the ninja who created the jutsu
2. ***JutsuName*** : varchar(50), part of composite primary key, foreign key referencing the jutsu that was created
3. creationDate : date, the date on which the jutsu was created

Rogue Association:

1. **ra_name** : varchar(50), primary key to uniquely identify each rogue organization
2. memberCount : integer, total number of members in the association
3. *leader* : integer, foreign key referencing the shinobi who leads the rogue association
4. formationDate : date, the date the rogue association was established
5. activityLevel : varchar(50), represents how active the association currently is (Low, Medium, High)
6. *originNationName* : varchar(50), foreign key referencing the village or nation where the association originated
7. objective : varchar(50), the primary goal or motivation of the rogue association

Ninja Mission:

1. **missionId** : integer, primary key to uniquely identify each mission
2. missionObjective : varchar(50), a brief description of what the mission entails
3. missionRank : char, the difficulty rank of the mission (D, C, B, A, S)
4. missionType : varchar(50), the category of the mission (Assassination, Escort, Recon etc.)
5. missionStatus : varchar(50), current state of the mission (Ongoing, Completed, Failed)
6. *teamAssigned* : integer, foreign key referencing the genin squad assigned to carry out the mission
7. *clientNation* : varchar(50), foreign key referencing the village or nation that requested the mission
8. revenue : decimal(10,2), the payment received upon mission completion

Standoff:

Design's Schema

1. ***ninjaId1*** : integer, part of composite primary key, foreign key referencing the first ninja involved in the standoff
2. ***ninjaId2*** : integer, part of composite primary key, foreign key referencing the second ninja involved in the standoff
3. Slocation : varchar(50), the location where the standoff took place
4. Sstatus : varchar(50), current status of the standoff (Ongoing, Completed)
5. *winnerNinjaId* : integer, foreign key referencing the ninja who won the standoff, nullable if ongoing

Chunin Competition:

1. **ChuninYear** : integer, primary key representing the year the competition was held
2. *winnerNinjaId* : integer, foreign key referencing the ninja who won the competition
3. *runnersUpNinjaId* : integer, foreign key referencing the ninja who came second
4. *venueNation* : varchar(50), foreign key referencing the village that hosted the competition
5. participantCount : integer, total number of ninjas who participated
6. *MainSponsorClan* : varchar(50), foreign key referencing the clan that sponsored the event
7. totalMatches : varchar(50), total number of matches held during the competition
8. duration : integer, number of days the competition lasted

Genin Squad:

1. **teamNo** : integer, primary key to uniquely identify each genin squad
2. *teamMentor* : integer, foreign key referencing the ninja who mentors the squad
3. *teamMember1* : integer, foreign key referencing the first member of the squad
4. *teamMember2* : integer, foreign key referencing the second member of the squad
5. *teamMember3* : integer, foreign key referencing the third member of the squad
6. SquadStatus : varchar(50), current status of the squad (Active, Disbanded etc.)
7. SquadRank : char, the overall rank of the squad
8. missionAccomplished : integer, total count of missions successfully completed by the squad

Inventory:

1. **ninjaGearId** : integer, primary key to uniquely identify each piece of ninja gear
2. Iname : varchar(50), the name of the inventory item
3. Irange : decimal(3,2), the effective range of the item in meters
4. worth : decimal(3,2), the monetary value of the item
5. Icount : integer, the quantity of this item available

6. rarity : varchar(50), the rarity tier of the item (Common, Rare, Legendary etc.)
7. *ownedByNation* : varchar(50), foreign key referencing the village that owns this item, nullable if unclaimed

Clan:

1. **clanName** : varchar(50), primary key to uniquely identify each clan
2. *originVillage* : varchar(50), foreign key referencing the village the clan originated from
3. specialAbilities : varchar(50), the unique power or kekkei genkai associated with the clan
4. clanSymbol : varchar(50), the symbol or emblem that represents the clan
5. clanSize : integer, the total number of members belonging to the clan

Led_By:

1. ***clanName*** : varchar(50), part of composite primary key, foreign key referencing the clan being led
2. ***NinjaId*** : integer, part of composite primary key, foreign key referencing the ninja who leads the clan
3. StartingDate : date, the date on which the ninja assumed leadership of the clan

Functionalities

1. Adding a new ninja to the Bingo Book

```
--1
--Adding a new ninja to the Bingo Book
Insert into shinobi (ninjaId,ninjaName,dateOfBirth,NinjaStatus,clan,chakraNature,jutsuName,village,ninjaRank,mentorNi
values(29,'Konohamaru Sarutobi','2007-09-24','Alive','Sarutobi','Wind','Rasengan','Leaf','C',7);

--2
```

```
Messages
11:01:38 AM   Started executing query at  Line 390
              (1 row affected)
              Total execution time: 00:00:00.117
```

2. Adding a new Village

```
o-Book-Shinobi-Intelligence-Database > Project.sql
--Adding a new Village
Insert into village (villageName,region,foundedYear,VillagePopulation,tailBeast,powerIndex)
values ('Sugukure','Land of Waterfalls',NULL, 3252000, '7-Tails', 7);
--3
```

```
Messages
11:03:05 AM   Started executing query at  Line 395
              (1 row affected)
              Total execution time: 00:00:00.115
```

3. Adding new standoff

```
--3
--Adding new standoff
Insert into standoff (ninjaId1, ninjaId2, Slocation, Sstatus, winnerNinjaId)
values (10, 4, 'Leaf', 'Completed', 4);
```

```
Messages
11:03:25 AM   Started executing query at  Line 400
              (1 row affected)
              Total execution time: 00:00:00.113
```

4. Adding new clan

```
--4
--Adding new clan
INSERT INTO clan (clanName, originVillage, specialAbilities, clanSymbol, clanSize) VALUES
('Uchihaa','Leaf','Sharingan','Fan',150);
```

```
Messages
11:03:55 AM   Started executing query at  Line 405
              (1 row affected)
              Total execution time: 00:00:00.114
```

5. Adding new Ninja Gear

```
--5
--Adding new Ninja Gear
INSERT INTO inventory (ninjaGearId, Iname, Irange, worth, Icount, rarity, ownedByNation) VALUES
(111,'Zabuza Executioner Blade',2.50,8.50,1,'Legendary',NULL);
```

```
Messages
11:04:13 AM   Started executing query at  Line 410
              (1 row affected)
              Total execution time: 00:00:00.113
```

6. Adding new Jutsu

```
--6
--Adding new Jutsu
INSERT INTO jutsu (jutsuName, jutsuType, coolDownTime, powerLevel,JutsuStatus,chakraConsumption,rankRequired,elementa
('Gentle Fisst','Taijutsu',3.00,7.5,'Clan Technique','Medium','B','None');
```

```
Messages
11:04:30 AM   Started executing query at  Line 415
              (1 row affected)
              Total execution time: 00:00:00.113
```

7. Adding new Rogue NinjaGang

```
--7
--Adding new Rogue NinjaGang
INSERT INTO rogueAssociation (ra_name, memberCount, leader, formationDate, activityLevel, originNationName, objective
('Akatsuki',10,21,'2005-01-01','High','Rain','World Peace via Pain');
```

```
Messages
11:04:42 AM   Started executing query at  Line 420
              (1 row affected)
              Total execution time: 00:00:00.115
```

8. Adding new log of chunin competition

Functionalities

```
--8
--Adding new log of chunin competition
INSERT INTO chuninCompetition (ChuninYear, winnerNinjaId, runnersUpNinjaId, venueNation, participantCount, MainSponso
(2001, 11, 17, 'Leaf', 30, 'Nara', '15', 7)

--SOME UPDATE STATEMETNS FUNCTIONALITY
```

```
Messages
11:04:54 AM   Started executing query at  Line 425
              (1 row affected)
              Total execution time: 00:00:00.114
```

9. Marking Death of a Ninja

```
--9
--Marking Death of a Ninja
Update shinobi        You, 9 hours ago • maybe now
set NinjaStatus = 'Deseased'
where ninjaId = 26;
```

```
11:05:04 AM   Started executing query at  Line 432
              (1 row affected)
              Total execution time: 00:00:00.068
```

10. Assign a Mentor to existing ninja

```
--10
--Assign a Mentor to existing ninja
Update shinobi
set mentorNinjaId = 14
where ninjaId = 17;
```

```
11:05:16 AM   Started executing query at  Line 438
              (1 row affected)
              Total execution time: 00:00:00.030
```

11. Update a jutsu with Cool Down Time equal to maximum cooldown time

```
--11
--update a jutsu with Cool Down Time equal to maximum cooldown time
update jutsu
set coolDownTime = 4.0
where coolDownTime =
    (
        select max(coolDownTime)
        from jutsu
    );
```

```
11:05:27 AM   Started executing query at  Line 444
              (1 row affected)
              Total execution time: 00:00:00.043
```

12. Retrieve shinobi count belonging to each village founded before 2000

```
--12
--Retrieve shinobi count belonging to each village founded before 2000
select v.villageName,count(*) as Shinobi_count
from village v
join shinobi s
on v.villageName = s.village
and v.foundedYear <= 2000
group by v.villageName;
```

	villageName	Shinobi_count
1	Cloud	1
2	Leaf	21
3	Rain	2
4	Sand	3
5	Stone	1

13. Retrieve shinobis

```
--Retreive shinobis
select * from village
--14. Update the rank of a ninja after promotion
UPDATE shinobi
SET ninjaRank = 'B'
WHERE ninjaId = 11;

--15. Change the leader of a rogue association
UPDATE rogueAssociation
SET leader = 22
WHERE ra_name = 'Taka';

--16. Delete all ninjas who have retired from the system
```

	villageName	region	foundedYear	VillagePopulation	tailBeast	powerl
1	Cloud	Land of Lightning	2000	9860	8-Tails	6.50
2	Leaf	Land of Fire	1937	1230	9-Tails	9.50
3	Mist	Land of Water	2005	5000	3-Tails	7.20
4	Rain	Land of Rivers	1999	1500	2-Tails	8.90
5	Sand	Land of Wind	1960	2560	1-Tail	7.80
6	Sound	Land of Fire	2007	1000	NULL	9.00
7	Stone	Land of Earth	1990	4350	4-Tails	8.10
8	Sugukure	Land of Waterfalls	NULL	3252000	7-Tails	7.00
9	Tagikakure	Land of Waterfalls	NULL	3252000	7-Tails	7.00

14. Update the rank of a ninja after promotion

Functionalities

```
--13
--Retreive shinobis
select * from village

--14. Update the rank of a ninja after promotion
UPDATE shinobi
SET ninjaRank = 'B'
WHERE ninjaId = 11;

--15. Change the leader of a rogue association
UPDATE rogueAssociation
SET leader = 22
WHERE ra_name = 'Taka';
```

```
11:06:27 AM   Started executing query at  Line 468
              (1 row affected)
              Total execution time: 00:00:00.105
```

15. Change the leader of a rogue association

```
--13
--Retreive shinobis
select * from village

--14. Update the rank of a ninja after promotion
UPDATE shinobi
SET ninjaRank = 'B'
WHERE ninjaId = 11;

--15. Change the leader of a rogue association
UPDATE rogueAssociation
SET leader = 22
WHERE ra_name = 'Taka';
```

```
11:06:37 AM   Started executing query at  Line 473
              (1 row affected)
              Total execution time: 00:00:00.113
```

16. Delete all ninjas who have retired from the system

```
--16. Delete all ninjas who have retired from the system
DELETE FROM shinobi
WHERE NinjaStatus = 'Retired';

--17. Remove a jutsu that is marked obsolete
DELETE FROM jutsu
WHERE jutsuStatus = 'Obsolete';

--18. Retrieve all ninjas belonging to a specific village
SELECT s.ninjaId, s.ninjaName, s.ninjaRank, s.chakraNature, s.clan
FROM shinobi s
WHERE s.village = 'Leaf'
ORDER BY s.ninjaRank, s.ninjaName;

--19. Find ninjas whose names start with a specific letter
```

```
Messages
11:08:05 AM   Started executing query at  Line 478
              (0 rows affected)
              Total execution time: 00:00:00.003
```

17. Remove a jutsu that is marked obsolete

18. Retrieve all ninjas belonging to a specific village

Functionalities

```sql
--18. Retrieve all ninjas belonging to a specific village
SELECT s.ninjaId, s.ninjaName, s.ninjaRank, s.chakraNature, s.clan
FROM shinobi s
WHERE s.village = 'Leaf'
ORDER BY s.ninjaRank, s.ninjaName;

--19. Find ninjas whose names start with a specific letter
SELECT ninjaId, ninjaName, village, ninjaRank
FROM shinobi
WHERE ninjaName LIKE 'S%'
ORDER BY ninjaName;

--20. Get all jutsu with power level greater than a given value
SELECT jutsuName, jutsuType, powerLevel, chakraConsumption, rankRequired
FROM jutsu
WHERE powerLevel > 8.0
ORDER BY powerLevel DESC;

--21. List villages with population above a threshold
SELECT villageName, region, VillagePopulation, powerIndex
FROM village
WHERE VillagePopulation > 500000
ORDER BY VillagePopulation DESC;

--22. Find all ninjas of a specific rank
SELECT ninjaId, ninjaName, village, clan, chakraNature
FROM shinobi
WHERE ninjaRank = 'S'
ORDER BY ninjaName;

--23. Count total number of ninjas in each village
SELECT v.villageName, COUNT(s.ninjaId) AS NinjaCount
```

Results (1) Messages

	ninjaId	ninjaName	ninjaRank	chakraNature	clan
1	11	Shikamaru Nara	B	Yin	Nara
2	13	Choji Akimichi	C	Yang	Akimichi
3	12	Ino Yamanaka	C	Yin	Yamanaka
4	28	Konohamaru Sarutobi	C	Wind	Sarutobi
5	29	Konohamaru Sarutobi	C	Wind	Sarutobi
6	27	Hinata Hyuga	G	Fire	Hyuga
7	7	Naruto Uzumaki	G	Wind	Uzumaki
8	26	Neji Hyuga	G	Wind	Hyuga
9	9	Sakura Haruno	G	Earth	NULL
10	8	Sasuke Uchiha	G	Lightning	Uchiha
11	1	Hiruzen Sarutobi	H	Fire	Sarutobi
12	5	Minato Namikaze	H	Wind	NULL
13	3	Tsunade Senju	H	Earth	Senju
14	10	Asuma Sarutobi	J	Wind	Sarutobi
15	6	Kakashi Hatake	J	Lightning	Hatake
16	14	Might Guy	J	Fire	NULL
17	15	Rock Lee	J	None	NULL
18	20	Itachi Uchiha	S	Fire	Uchiha
19	2	Jiraiya	S	Fire	NULL
20	4	Orochimaru	S	Wind	NULL
21	23	Tobi	S	None	Uchiha

19. Find ninjas whose names start with a specific letter

```sql
--18. Retrieve all ninjas belonging to a specific village
SELECT s.ninjaId, s.ninjaName, s.ninjaRank, s.chakraNature, s.clan
FROM shinobi s
WHERE s.village = 'Leaf'
ORDER BY s.ninjaRank, s.ninjaName;

--19. Find ninjas whose names start with a specific letter
SELECT ninjaId, ninjaName, village, ninjaRank
FROM shinobi
WHERE ninjaName LIKE 'S%'
ORDER BY ninjaName;

--20. Get all jutsu with power level greater than a given value
SELECT jutsuName, jutsuType, powerLevel, chakraConsumption, rankRequired
FROM jutsu
WHERE powerLevel > 8.0
ORDER BY powerLevel DESC;

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SELECT villageName, region, VillagePopulation, powerIndex
FROM village
WHERE VillagePopulation > 500000
ORDER BY VillagePopulation DESC;

--22. Find all ninjas of a specific rank
SELECT ninjaId, ninjaName, village, clan, chakraNature
FROM shinobi
WHERE ninjaRank = 'S'
ORDER BY ninjaName;

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SELECT v.villageName, COUNT(s.ninjaId) AS NinjaCount
```

Results (1) Messages

	ninjaId	ninjaName	village	ninjaRank
1	9	Sakura Haruno	Leaf	G
2	8	Sasuke Uchiha	Leaf	G
3	11	Shikamaru Nara	Leaf	B

20. Get all jutsu with power level greater than a given value

Functionalities

```sql
--18. Retrieve all ninjas belonging to a specific village
SELECT s.ninjaId, s.ninjaName, s.ninjaRank, s.chakraNature, s.clan
FROM shinobi s
WHERE s.village = 'Leaf'
ORDER BY s.ninjaRank, s.ninjaName;

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FROM shinobi
WHERE ninjaRank = 'S'
ORDER BY ninjaName;

--23. Count total number of ninjas in each village
```

	jutsuName	jutsuType	powerLevel	chakraConsumption	rankRequired
1	Almighty Push	Dojutsu	9.90	Extreme	S
2	Amaterasu	Dojutsu	9.90	Extreme	S
3	Kamui	Dojutsu	9.70	Extreme	S
4	Kirin	Ninjutsu	9.50	Extreme	S
5	Daytime Tiger	Taijutsu	9.20	Extreme	S
6	C4 Karura	Ninjutsu	9.10	Extreme	S
7	Hidden Lotus	Taijutsu	8.80	Extreme	A
8	Lariat	Taijutsu	8.70	High	A
9	Rasengan	Ninjutsu	8.50	High	A
10	Snake Authority	Ninjutsu	8.20	High	A

21. List villages with population above a threshold

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--18. Retrieve all ninjas belonging to a specific village
SELECT s.ninjaId, s.ninjaName, s.ninjaRank, s.chakraNature, s.clan
FROM shinobi s
WHERE s.village = 'Leaf'
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FROM shinobi
WHERE ninjaRank = 'S'
ORDER BY ninjaName;

--23. Count total number of ninjas in each village
```

Results (1) Messages

	villageName	region	VillagePopulation	powerIndex
1	Sugukure	Land of Waterfalls	3252000	7.00
2	Tagikakure	Land of Waterfalls	3252000	7.00

22. Find all ninjas of a specific rank

Functionalities

```sql
--18. Retrieve all ninjas belonging to a specific village
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FROM shinobi
WHERE ninjaRank = 'S'
ORDER BY ninjaName;

--23. Count total number of ninjas in each village
SELECT v.villageName, COUNT(s.ninjaId) AS NinjaCount
```

	ninjaId	ninjaName	village	clan	chakraNature
1	24	Deidara	Stone	NULL	Earth
2	20	Itachi Uchiha	Leaf	Uchiha	Fire
3	2	Jiraiya	Leaf	NULL	Fire
4	25	Kisame Hoshigaki	Mist	Hoshigaki	Water
5	22	Konan	Rain	NULL	None
6	4	Orochimaru	Leaf	NULL	Wind
7	21	Pain	Rain	NULL	None
8	23	Tobi	Leaf	Uchiha	None

23. Count total number of ninjas in each village

```sql
--23. Count total number of ninjas in each village
SELECT v.villageName, COUNT(s.ninjaId) AS NinjaCount
FROM village v
LEFT JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
ORDER BY NinjaCount DESC;

--24. Find ninja ranks with average chakra consumption above a threshold
SELECT
    ninjaRank,
    COUNT(*) AS NinjaCount,
    AVG(CASE j.chakraConsumption
            WHEN 'Low'     THEN 1
            WHEN 'Medium'  THEN 2
            WHEN 'High'    THEN 3
            WHEN 'Extreme' THEN 4
            ELSE 0
        END) AS AvgChakraWeight
FROM shinobi s
LEFT JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY ninjaRank
HAVING AVG  (CASE j.chakraConsumption
            WHEN 'Low'     THEN 1
            WHEN 'Medium'  THEN 2
            WHEN 'High'    THEN 3
            WHEN 'Extreme' THEN 4
            ELSE 0
           END) > 2
ORDER BY AvgChakraWeight DESC;

--25. Retrieve ninja names along with their village details
```

Results (1) Messages

	villageName	NinjaCount
1	Leaf	21
2	Sand	3
3	Rain	2
4	Cloud	1
5	Mist	1
6	Stone	1
7	Sugukure	0
8	Tagikakure	0
9	Sound	0

24. Find ninja ranks with average chakra consumption above a threshold

Functionalities

```sql
--23. Count total number of ninjas in each village
SELECT v.villageName, COUNT(s.ninjaId) AS NinjaCount
FROM village v
LEFT JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
ORDER BY NinjaCount DESC;

--24. Find ninja ranks with average chakra consumption above a threshold
SELECT
    ninjaRank,
    COUNT(*) AS NinjaCount,
    AVG(CASE j.chakraConsumption
                WHEN 'Low'     THEN 1
                WHEN 'Medium'  THEN 2
                WHEN 'High'    THEN 3
                WHEN 'Extreme' THEN 4
                ELSE 0
        END) AS AvgChakraWeight
FROM shinobi s
LEFT JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY ninjaRank
HAVING AVG  (CASE j.chakraConsumption
                WHEN 'Low'     THEN 1
                WHEN 'Medium'  THEN 2
                WHEN 'High'    THEN 3
                WHEN 'Extreme' THEN 4
                ELSE 0
            END) > 2
ORDER BY AvgChakraWeight DESC;

--25. Retrieve ninja names along with their village details
```

Results (1) Messages

	ninjaRank	NinjaCount	AvgChakraWeight
1	J	7	3
2	S	8	3

25. Retrieve ninja names along with their village details

```sql
--24. Find ninja ranks with average chakra consumption above a threshold
SELECT
    ninjaRank,
    COUNT(*) AS NinjaCount,
    AVG(CASE j.chakraConsumption
                WHEN 'Low'     THEN 1
                WHEN 'Medium'  THEN 2
                WHEN 'High'    THEN 3
                WHEN 'Extreme' THEN 4
                ELSE 0
        END) AS AvgChakraWeight
FROM shinobi s
LEFT JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY ninjaRank
HAVING AVG  (CASE j.chakraConsumption
                WHEN 'Low'     THEN 1
                WHEN 'Medium'  THEN 2
                WHEN 'High'    THEN 3
                WHEN 'Extreme' THEN 4
                ELSE 0
            END) > 2
ORDER BY AvgChakraWeight DESC;

--25. Retrieve ninja names along with their village details
SELECT s.ninjaId, s.ninjaName, s.ninjaRank,
    v.villageName, v.region, v.powerIndex
FROM shinobi s
INNER JOIN village v ON s.village = v.villageName;

--26. List all jutsu along with the ninja who created them
SELECT j.jutsuName, j.jutsuType, j.powerLevel,
    s.ninjaName AS CreatedBy, cb.creationDate
```

Results (1) Messages

	ninjaId	ninjaName	ninjaRank	villageName	region	powerIndex
1	1	Hiruzen Sarutobi	H	Leaf	Land of Fire	9.50
2	2	Jiraiya	S	Leaf	Land of Fire	9.50
3	3	Tsunade Senju	H	Leaf	Land of Fire	9.50
4	4	Orochimaru	S	Leaf	Land of Fire	9.50
5	5	Minato Namikaze	H	Leaf	Land of Fire	9.50
6	6	Kakashi Hatake	J	Leaf	Land of Fire	9.50
7	7	Naruto Uzumaki	G	Leaf	Land of Fire	9.50
8	8	Sasuke Uchiha	G	Leaf	Land of Fire	9.50
9	9	Sakura Haruno	G	Leaf	Land of Fire	9.50
10	10	Asuma Sarutobi	J	Leaf	Land of Fire	9.50
11	11	Shikamaru Nara	B	Leaf	Land of Fire	9.50
12	12	Ino Yamanaka	C	Leaf	Land of Fire	9.50
13	13	Choji Akimichi	C	Leaf	Land of Fire	9.50
14	14	Might Guy	J	Leaf	Land of Fire	9.50
15	15	Rock Lee	J	Leaf	Land of Fire	9.50
16	16	Gaara	K	Sand	Land of Wind	7.80
17	17	Temari	J	Sand	Land of Wind	7.80
18	18	Kankuro	J	Sand	Land of Wind	7.80
19	19	Killer Bee	J	Cloud	Land of Lightning	6.50
20	20	Itachi Uchiha	S	Leaf	Land of Fire	9.50
21	21	Pain	S	Rain	Land of Rivers	8.90
22	22	Konan	S	Rain	Land of Rivers	8.90

26. List all jutsu along with the ninja who created them

Functionalities

```sql
INNER JOIN village v ON s.village = v.villageName;

--26. List all jutsu along with the ninja who created them
SELECT j.jutsuName, j.jutsuType, j.powerLevel,
       s.ninjaName AS CreatedBy, cb.creationDate
FROM jutsu j
INNER JOIN created_by cb ON j.jutsuName = cb.JutsuName
INNER JOIN shinobi s     ON cb.NinjaID  = s.ninjaId;
You, 9 hours ago • maybe now
--27. Display missions along with the squad assigned to them
SELECT m.missionId, m.missionObjective, m.missionRank, m.missionStatus, m.revenue,
       g.teamNo, g.SquadRank, g.missionAccomplished
FROM ninjaMission m
INNER JOIN geninSquad g ON m.teamAssigned = g.teamNo;

--28. List all ninjas and their mentors, including those without mentors
SELECT s.ninjaId, s.ninjaName, s.ninjaRank,
       m.ninjaName AS MentorName, m.ninjaRank AS MentorRank
FROM shinobi s
LEFT JOIN shinobi m ON s.mentorNinjaId = m.ninjaId
ORDER BY s.ninjaName;

--29. Find ninjas who are both clan members and mission participants
SELECT DISTINCT s.ninjaId, s.ninjaName, s.clan, s.village
FROM shinobi s
INNER JOIN geninSquad g ON (s.ninjaId = g.teamMember1
                        OR s.ninjaId = g.teamMember2
                        OR s.ninjaId = g.teamMember3)
INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned
WHERE s.clan IS NOT NULL;

--30. Get ninjas who have never participated in any mission
```

Results (1) Messages Open in New Tab

	jutsuName	jutsuType	powerLevel	CreatedBy	creationDate
1	Fire Dragon Flame	Ninjutsu	7.50	Hiruzen Sarutobi	1995-01-01
2	Heavenly Foot of Pain	Taijutsu	7.00	Tsunade Senju	2006-01-01
3	Snake Authority	Ninjutsu	8.20	Orochimaru	1998-01-01
4	Rasengan	Ninjutsu	8.50	Minato Namikaze	2001-01-01
5	Chidori	Ninjutsu	8.00	Kakashi Hatake	2003-01-01
6	Kirin	Ninjutsu	9.50	Sasuke Uchiha	2009-01-01
7	Shadow Stitching	Hiden	6.50	Shikamaru Nara	2007-01-01
8	Mind Transfer	Hiden	6.00	Ino Yamanaka	2007-01-01
9	Human Bullet Tank	Taijutsu	6.80	Choji Akimichi	2007-01-01
10	Daytime Tiger	Taijutsu	9.20	Might Guy	2005-01-01
11	Hidden Lotus	Taijutsu	8.80	Might Guy	2003-01-01
12	Sand Coffin	Ninjutsu	7.90	Gaara	2004-01-01
13	Sea Dragon	Ninjutsu	7.40	Temari	2006-01-01
14	Lariat	Taijutsu	8.70	Killer Bee	2002-01-01
15	Amaterasu	Dojutsu	9.90	Itachi Uchiha	2008-01-01
16	Almighty Push	Dojutsu	9.90	Pain	2006-01-01
17	Paper Shuriken	Ninjutsu	7.20	Konan	2007-01-01
18	Kamui	Dojutsu	9.70	Tobi	2005-01-01
19	C4 Karura	Ninjutsu	9.10	Deidara	2008-01-01
20	Water Dragon	Ninjutsu	7.80	Kisame Hoshigaki	2003-01-01
21	Gentle Fist	Taijutsu	7.50	Neji Hyuga	2007-01-01

27. Display missions along with the squad assigned to them

```sql
--25. Retrieve ninja names along with their village details
SELECT s.ninjaId, s.ninjaName, s.ninjaRank,
       v.villageName, v.region, v.powerIndex
FROM shinobi s
INNER JOIN village v ON s.village = v.villageName;

--26. List all jutsu along with the ninja who created them
SELECT j.jutsuName, j.jutsuType, j.powerLevel,
       s.ninjaName AS CreatedBy, cb.creationDate
FROM jutsu j
INNER JOIN created_by cb ON j.jutsuName = cb.JutsuName
INNER JOIN shinobi s     ON cb.NinjaID  = s.ninjaId;

--27. Display missions along with the squad assigned to them
SELECT m.missionId, m.missionObjective, m.missionRank, m.missionStatus, m.revenue,      You, 9 hours ago • maybe now
       g.teamNo, g.SquadRank, g.missionAccomplished
FROM ninjaMission m
INNER JOIN geninSquad g ON m.teamAssigned = g.teamNo;

--28. List all ninjas and their mentors, including those without mentors
SELECT s.ninjaId, s.ninjaName, s.ninjaRank,
       m.ninjaName AS MentorName, m.ninjaRank AS MentorRank
FROM shinobi s
LEFT JOIN shinobi m ON s.mentorNinjaId = m.ninjaId
ORDER BY s.ninjaName;

--29. Find ninjas who are both clan members and mission participants
SELECT DISTINCT s.ninjaId, s.ninjaName, s.clan, s.village
FROM shinobi s
INNER JOIN geninSquad g ON (s.ninjaId = g.teamMember1
                        OR s.ninjaId = g.teamMember2
                        OR s.ninjaId = g.teamMember3)
```

Results (1) Messages Open in New Tab

	missionId	missionObjective	missionRank	missionStatus	revenue	tea
1	101	Escort Bridge Builder Tazuna	A	NULL	50000.00	7
2	102	Retrieve Forbidden Scroll	S	NULL	150000.00	7
3	103	Infiltrate Hidden Rain	S	NULL	200000.00	1
4	104	Defend Sunagakure Borders	B	NULL	35000.00	5
5	105	Capture Rare Medicinal Herbs	C	NULL	12000.00	10
6	106	Protect Land of Fire Daimyo	A	NULL	80000.00	9
7	107	Investigate Rogue Activity	B	NULL	45000.00	10
8	108	Intercept Akatsuki Spy	S	NULL	180000.00	13
9	109	Deliver Secret Intelligence	B	NULL	30000.00	8
10	110	Search for Rare Ore	C	NULL	15000.00	12
11	111	Escort Merchant Caravan	C	NULL	10000.00	8
12	112	Neutralize Bandit Leader	B	NULL	25000.00	9
13	113	Reconnaissance in Land of Iron	A	NULL	70000.00	5
14	114	Sabotage Weapon Supply Line	A	NULL	90000.00	13
15	115	Recover Stolen Kunai Shipment	D	NULL	5000.00	7

28. List all ninjas and their mentors, including those without mentors

Functionalities

```sql
--25. Retrieve ninja names along with their village details
SELECT s.ninjaId, s.ninjaName, s.ninjaRank,
       v.villageName, v.region, v.powerIndex
FROM shinobi s
INNER JOIN village v ON s.village = v.villageName;

--26. List all jutsu along with the ninja who created them
SELECT j.jutsuName, j.jutsuType, j.powerLevel,
       s.ninjaName AS CreatedBy, cb.creationDate
FROM jutsu j
INNER JOIN created_by cb ON j.jutsuName = cb.JutsuName
INNER JOIN shinobi s     ON cb.NinjaID  = s.ninjaId;

--27. Display missions along with the squad assigned to them
SELECT m.missionId, m.missionObjective, m.missionRank, m.missionStatus, m.revenue,
       g.teamNo, g.SquadRank, g.missionAccomplished
FROM ninjaMission m
INNER JOIN geninSquad g ON m.teamAssigned = g.teamNo;

--28. List all ninjas and their mentors, including those without mentors
SELECT s.ninjaId, s.ninjaName, s.ninjaRank,
       m.ninjaName AS MentorName, m.ninjaRank AS MentorRank
FROM shinobi s
LEFT JOIN shinobi m ON s.mentorNinjaId = m.ninjaId
ORDER BY s.ninjaName;

--29. Find ninjas who are both clan members and mission participants
SELECT DISTINCT s.ninjaId, s.ninjaName, s.clan, s.village
FROM shinobi s
INNER JOIN geninSquad g ON (s.ninjaId = g.teamMember1
                            OR s.ninjaId = g.teamMember2
                            OR s.ninjaId = g.teamMember3)
```

Results (1) Messages Open in New Tab

	ninjaId	ninjaName	ninjaRank	MentorName	MentorRank
1	10	Asuma Sarutobi	J	Hiruzen Sarutobi	H
2	13	Choji Akimichi	C	Asuma Sarutobi	J
3	24	Deidara	S	NULL	NULL
4	16	Gaara	K	NULL	NULL
5	27	Hinata Hyuga	G	Hiruzen Sarutobi	H
6	1	Hiruzen Sarutobi	H	NULL	NULL
7	12	Ino Yamanaka	C	Asuma Sarutobi	J
8	20	Itachi Uchiha	S	Orochimaru	S
9	2	Jiraiya	S	Hiruzen Sarutobi	H
10	6	Kakashi Hatake	J	Minato Namikaze	H
11	18	Kankuro	J	NULL	NULL
12	19	Killer Bee	J	NULL	NULL
13	25	Kisame Hoshigaki	S	NULL	NULL
14	22	Konan	S	Jiraiya	S
15	28	Konohamaru Sarutobi	C	Naruto Uzumaki	G
16	29	Konohamaru Sarutobi	C	Naruto Uzumaki	G
17	14	Might Guy	J	NULL	NULL
18	5	Minato Namikaze	H	Jiraiya	S
19	7	Naruto Uzumaki	G	Kakashi Hatake	J
20	26	Neji Hyuga	G	Might Guy	J
21	4	Orochimaru	S	Hiruzen Sarutobi	H
22	21	Pain	S	Jiraiya	S

29. Find ninjas who are both clan members and mission participants

```sql
LEFT JOIN shinobi m ON s.mentorNinjaId = m.ninjaId
ORDER BY s.ninjaName;

--29. Find ninjas who are both clan members and mission participants
SELECT DISTINCT s.ninjaId, s.ninjaName, s.clan, s.village
FROM shinobi s
INNER JOIN geninSquad g ON (s.ninjaId = g.teamMember1
                            OR s.ninjaId = g.teamMember2
                            OR s.ninjaId = g.teamMember3)
INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned
WHERE s.clan IS NOT NULL;

--30. Get ninjas who have never participated in any mission
SELECT s.ninjaId, s.ninjaName, s.village, s.ninjaRank
FROM shinobi s
WHERE s.ninjaId NOT IN (
    SELECT DISTINCT g.teamMember1 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
    UNION
    SELECT DISTINCT g.teamMember2 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
    UNION
    SELECT DISTINCT g.teamMember3 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
)
ORDER BY s.ninjaName;

--31. Find the clan with the maximum number of members
SELECT TOP 1 clanName, clanSize, originVillage, specialAbilities
FROM clan
ORDER BY clanSize DESC;

--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
```

Results (1) Messages Open in New Tab

	ninjaId	ninjaName	clan	village
1	3	Tsunade Senju	Senju	Leaf
2	7	Naruto Uzumaki	Uzumaki	Leaf
3	8	Sasuke Uchiha	Uchiha	Leaf
4	11	Shikamaru Nara	Nara	Leaf
5	12	Ino Yamanaka	Yamanaka	Leaf
6	13	Choji Akimichi	Akimichi	Leaf
7	23	Tobi	Uchiha	Leaf
8	25	Kisame Hoshigaki	Hoshigaki	Mist

30. Get ninjas who have never participated in any mission

Functionalities

```sql
LEFT JOIN shinobi m ON s.mentorNinjaId = m.ninjaId
ORDER BY s.ninjaName;

--29. Find ninjas who are both clan members and mission participants
SELECT DISTINCT s.ninjaId, s.ninjaName, s.clan, s.village
FROM shinobi s
INNER JOIN geninSquad g ON (s.ninjaId = g.teamMember1
                            OR s.ninjaId = g.teamMember2
                            OR s.ninjaId = g.teamMember3)
INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned
WHERE s.clan IS NOT NULL;

--30. Get ninjas who have never participated in any mission
SELECT s.ninjaId, s.ninjaName, s.village, s.ninjaRank
FROM shinobi s
WHERE s.ninjaId NOT IN (
    SELECT DISTINCT g.teamMember1 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
    UNION
    SELECT DISTINCT g.teamMember2 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
    UNION
    SELECT DISTINCT g.teamMember3 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
)
ORDER BY s.ninjaName;
You, 9 hours ago • maybe now
--31. Find the clan with the maximum number of members
SELECT TOP 1 clanName, clanSize, originVillage, specialAbilities
FROM clan
ORDER BY clanSize DESC;

--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
```

Open in New Tab

	ninjaId	ninjaName	village	ninjaRank
1	10	Asuma Sarutobi	Leaf	J
2	27	Hinata Hyuga	Leaf	G
3	1	Hiruzen Sarutobi	Leaf	H
4	20	Itachi Uchiha	Leaf	S
5	6	Kakashi Hatake	Leaf	J
6	28	Konohamaru Sarutobi	Leaf	C
7	29	Konohamaru Sarutobi	Leaf	C
8	14	Might Guy	Leaf	J
9	5	Minato Namikaze	Leaf	H
10	26	Neji Hyuga	Leaf	G

31. Find the clan with the maximum number of members

```sql
LEFT JOIN shinobi m ON s.mentorNinjaId = m.ninjaId
ORDER BY s.ninjaName;

--29. Find ninjas who are both clan members and mission participants
SELECT DISTINCT s.ninjaId, s.ninjaName, s.clan, s.village
FROM shinobi s
INNER JOIN geninSquad g ON (s.ninjaId = g.teamMember1
                            OR s.ninjaId = g.teamMember2
                            OR s.ninjaId = g.teamMember3)
INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned
WHERE s.clan IS NOT NULL;

--30. Get ninjas who have never participated in any mission
SELECT s.ninjaId, s.ninjaName, s.village, s.ninjaRank
FROM shinobi s
WHERE s.ninjaId NOT IN (
    SELECT DISTINCT g.teamMember1 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
    UNION
    SELECT DISTINCT g.teamMember2 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
    UNION
    SELECT DISTINCT g.teamMember3 FROM geninSquad g INNER JOIN ninjaMission m ON g.teamNo = m.teamAssigned WHERE g.tea
)
ORDER BY s.ninjaName;

--31. Find the clan with the maximum number of members
SELECT TOP 1 clanName, clanSize, originVillage, specialAbilities
FROM clan
ORDER BY clanSize DESC;
You, 9 hours ago • maybe now
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
```

Results (1) Messages

Open in New Tab

	clanName	clanSize	originVillage	specialAbilities
1	Uchiha	150	Leaf	Sharingan

32. AVG powerlevel

Functionalities

```
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
FROM jutsu;
      You, 9 hours ago • maybe now
--33.Max revenue
SELECT MAX(revenue) AS MAX_Reveneue
FROM ninjaMission;

--34.Villages sorted by power index

SELECT villagename,powerIndex
FROM village
ORDER BY powerIndex;

--35.Sum of mission revenue
SELECT SUM(revenue) AS TOTAL_Reveneue
FROM ninjaMission;

--36. Average combat strength
SELECT s.village AS VillageName, AVG(j.powerLevel) AS AvgNinjaStrength, COUNT(s.ninjaId) AS TotalNinjas
FROM shinobi s
JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY s.village
HAVING COUNT(s.ninjaId) > 1
ORDER BY AvgNinjaStrength DESC;

--37. Shinobis sorted by DOB
SELECT ninjaName,dateOfBirth
FROM shinobi
ORDER BY dateOfBirth;

--38. list of villages along with the total number of ninjas residing in each
```

Results (1) Messages Open in New Tab

	AVG_JUSTU
1	8.118181

33. Max revenue

```
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
FROM jutsu;

--33.Max revenue
SELECT MAX(revenue) AS MAX_Reveneue
FROM ninjaMission;
      You, 9 hours ago • maybe now
--34.Villages sorted by power index

SELECT villagename,powerIndex
FROM village
ORDER BY powerIndex;

--35.Sum of mission revenue
SELECT SUM(revenue) AS TOTAL_Reveneue
FROM ninjaMission;

--36. Average combat strength
SELECT s.village AS VillageName, AVG(j.powerLevel) AS AvgNinjaStrength, COUNT(s.ninjaId) AS TotalNinjas
FROM shinobi s
JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY s.village
HAVING COUNT(s.ninjaId) > 1
ORDER BY AvgNinjaStrength DESC;

--37. Shinobis sorted by DOB
SELECT ninjaName,dateOfBirth
FROM shinobi
ORDER BY dateOfBirth;

--38. list of villages along with the total number of ninjas residing in each
```

Open in New Tab

	MAX_Reveneue
1	200000.00

34. Villages sorted by power index

Functionalities

```
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
FROM jutsu;

--33.Max revenue
SELECT MAX(revenue) AS MAX_Reveneue
FROM ninjaMission;

--34.Villages sorted by power index

SELECT villagename,powerIndex
FROM village
ORDER BY powerIndex;

--35.Sum of mission revenue      You, 9 hours ago • maybe now
SELECT SUM(revenue) AS TOTAL_Reveneue
FROM ninjaMission;

--36. Average combat strength
SELECT s.village AS VillageName, AVG(j.powerLevel) AS AvgNinjaStrength, COUNT(s.ninjaId) AS TotalNinjas
FROM shinobi s
JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY s.village
HAVING COUNT(s.ninjaId) > 1
ORDER BY AvgNinjaStrength DESC;

--37. Shinobis sorted by DOB
SELECT ninjaName,dateOfBirth
FROM shinobi
ORDER BY dateOfBirth;

--38. list of villages along with the total number of ninjas residing in each
```

	villagename	powerIndex
1	Cloud	6.50
2	Sugukure	7.00
3	Tagikakure	7.00
4	Mist	7.20
5	Sand	7.80
6	Stone	8.10
7	Rain	8.90
8	Sound	9.00
9	Leaf	9.50

35. Sum of mission revenue

```
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
FROM jutsu;

--33.Max revenue
SELECT MAX(revenue) AS MAX_Reveneue
FROM ninjaMission;

--34.Villages sorted by power index

SELECT villagename,powerIndex
FROM village
ORDER BY powerIndex;

--35.Sum of mission revenue
SELECT SUM(revenue) AS TOTAL_Reveneue
FROM ninjaMission;
      You, 9 hours ago • maybe now
--36. Average combat strength
SELECT s.village AS VillageName, AVG(j.powerLevel) AS AvgNinjaStrength, COUNT(s.ninjaId) AS TotalNinjas
FROM shinobi s
JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY s.village
HAVING COUNT(s.ninjaId) > 1
ORDER BY AvgNinjaStrength DESC;

--37. Shinobis sorted by DOB
SELECT ninjaName,dateOfBirth
FROM shinobi
ORDER BY dateOfBirth;

--38. list of villages along with the total number of ninjas residing in each
```

	TOTAL_Reveneue
1	997000.00

36. Average combat strength

Functionalities

```sql
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
FROM jutsu;

--33.Max revenue
SELECT MAX(revenue) AS MAX_Reveneue
FROM ninjaMission;

--34.Villages sorted by power index

SELECT villagename,powerIndex
FROM village
ORDER BY powerIndex;

--35.Sum of mission revenue
SELECT SUM(revenue) AS TOTAL_Reveneue
FROM ninjaMission;

--36. Average combat strength
SELECT s.village AS VillageName, AVG(j.powerLevel) AS AvgNinjaStrength, COUNT(s.ninjaId) AS TotalNinjas
FROM shinobi s
JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY s.village
HAVING COUNT(s.ninjaId) > 1
ORDER BY AvgNinjaStrength DESC;
      You, 9 hours ago • maybe now
--37. Shinobis sorted by DOB
SELECT ninjaName,dateOfBirth
FROM shinobi
ORDER BY dateOfBirth;

--38. list of villages along with the total number of ninjas residing in each
```

Open in New Tab

	VillageName	AvgNinjaStrength	TotalNinjas
1	Rain	8.550000	2
2	Leaf	8.052380	21
3	Sand	7.733333	3

37. Shinobis sorted by DOB

```sql
--32.AVG powerlevel
SELECT AVG(powerLevel) as AVG_JUSTU
FROM jutsu;

--33.Max revenue
SELECT MAX(revenue) AS MAX_Reveneue
FROM ninjaMission;

--34.Villages sorted by power index

SELECT villagename,powerIndex
FROM village
ORDER BY powerIndex;

--35.Sum of mission revenue
SELECT SUM(revenue) AS TOTAL_Reveneue
FROM ninjaMission;

--36. Average combat strength
SELECT s.village AS VillageName, AVG(j.powerLevel) AS AvgNinjaStrength, COUNT(s.ninjaId) AS TotalNinjas
FROM shinobi s
JOIN jutsu j ON s.jutsuName = j.jutsuName
GROUP BY s.village
HAVING COUNT(s.ninjaId) > 1
ORDER BY AvgNinjaStrength DESC;

--37. Shinobis sorted by DOB
SELECT ninjaName,dateOfBirth
FROM shinobi
ORDER BY dateOfBirth;
      You, 9 hours ago • maybe now
--38. list of villages along with the total number of ninjas residing in each
```

Open in New Tab

	ninjaName	dateOfBirth
1	Hiruzen Sarutobi	1937-02-08
2	Jiraiya	1946-11-11
3	Tsunade Senju	1950-08-02
4	Orochimaru	1951-10-27
5	Minato Namikaze	1968-01-25
6	Might Guy	1970-01-01
7	Killer Bee	1970-05-15
8	Asuma Sarutobi	1970-10-18
9	Kakashi Hatake	1972-09-15
10	Kisame Hoshigaki	1980-03-18
11	Tobi	1983-02-10
12	Itachi Uchiha	1985-06-09
13	Konan	1987-02-20
14	Pain	1987-09-19
15	Deidara	1988-05-05
16	Temari	1997-08-23
17	Kankuro	1998-05-15
18	Gaara	1999-01-19
19	Sakura Haruno	1999-03-28
20	Choji Akimichi	1999-05-01
21	Neji Hyuga	1999-07-03
22	Sasuke Uchiha	1999-07-23

38. List of villages along with the total number of ninjas residing in each

Functionalities

```
--38. list of villages along with the total number of ninjas residing in each
SELECT v.villageName,COUNT(s.ninjaId)
FROM village v
JOIN shinobi s on s.village=v.villageName
GROUP BY v.villageName
HAVING COUNT(s.ninjaId)>1
      You, 9 hours ago • maybe now
--39. Jutsu and their creators
SELECT s.NinjaID,j.jutsuName,j.jutsuType AS Jutsu_type,s.ninjaName AS Creator_name ,j.powerLevel
FROM shinobi s
JOIN created_by c ON s.ninjaId = c.NinjaID
JOIN jutsu j ON c.JutsuName=j.jutsuName;

--40.Clan and their leaders
SELECT l.NinjaId,c.clanName,s.ninjaName AS leader_name
FROM clan c
JOIN led_by l ON c.clanName=l.clanName
JOIN shinobi s on s.ninjaId=l.NinjaId;

--41.Ninjas from 2 specific villages
SELECT ninjaName,village
FROM shinobi
WHERE village = 'Leaf '
   OR village = 'Rain ';

--42.Villages having more ninjas than average
SELECT v.villageName,COUNT(s.ninjaId) AS NinjaCount
FROM village v
JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
HAVING COUNT(s.ninjaId) > (
    SELECT AVG(VillageCounts.Total)
```

	villageName	(No column name)
1	Leaf	21
2	Rain	2
3	Sand	3

39. Jutsu and their creators

```
--38. list of villages along with the total number of ninjas residing in each
SELECT v.villageName,COUNT(s.ninjaId)
FROM village v
JOIN shinobi s on s.village=v.villageName
GROUP BY v.villageName
HAVING COUNT(s.ninjaId)>1

--39. Jutsu and their creators
SELECT s.NinjaID,j.jutsuName,j.jutsuType AS Jutsu_type,s.ninjaName AS Creator_name ,j.powerLevel
FROM shinobi s
JOIN created_by c ON s.ninjaId = c.NinjaID
JOIN jutsu j ON c.JutsuName=j.jutsuName;
      You, 9 hours ago • maybe now
--40.Clan and their leaders
SELECT l.NinjaId,c.clanName,s.ninjaName AS leader_name
FROM clan c
JOIN led_by l ON c.clanName=l.clanName
JOIN shinobi s on s.ninjaId=l.NinjaId;

--41.Ninjas from 2 specific villages
SELECT ninjaName,village
FROM shinobi
WHERE village = 'Leaf '
   OR village = 'Rain ';

--42.Villages having more ninjas than average
SELECT v.villageName,COUNT(s.ninjaId) AS NinjaCount
FROM village v
JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
HAVING COUNT(s.ninjaId) > (
    SELECT AVG(VillageCounts.Total)
```

	NinjaID	jutsuName	Jutsu_type	Creator_name	powerLevel
1	1	Fire Dragon Flame	Ninjutsu	Hiruzen Sarutobi	7.50
2	3	Heavenly Foot of Pain	Taijutsu	Tsunade Senju	7.00
3	4	Snake Authority	Ninjutsu	Orochimaru	8.20
4	5	Rasengan	Ninjutsu	Minato Namikaze	8.50
5	6	Chidori	Ninjutsu	Kakashi Hatake	8.00
6	8	Kirin	Ninjutsu	Sasuke Uchiha	9.50
7	11	Shadow Stitching	Hiden	Shikamaru Nara	6.50
8	12	Mind Transfer	Hiden	Ino Yamanaka	6.00
9	13	Human Bullet Tank	Taijutsu	Choji Akimichi	6.80
10	14	Daytime Tiger	Taijutsu	Might Guy	9.20
11	14	Hidden Lotus	Taijutsu	Might Guy	8.80
12	16	Sand Coffin	Ninjutsu	Gaara	7.90
13	17	Sea Dragon	Ninjutsu	Temari	7.40
14	19	Lariat	Taijutsu	Killer Bee	8.70
15	20	Amaterasu	Dojutsu	Itachi Uchiha	9.90
16	21	Almighty Push	Dojutsu	Pain	9.90
17	22	Paper Shuriken	Ninjutsu	Konan	7.20
18	23	Kamui	Dojutsu	Tobi	9.70
19	24	C4 Karura	Ninjutsu	Deidara	9.10
20	25	Water Dragon	Ninjutsu	Kisame Hoshigaki	7.80
21	26	Gentle Fist	Taijutsu	Neji Hyuga	7.50

40. Clan and their leaders

Functionalities

```sql
--38. list of villages along with the total number of ninjas residing in each
SELECT v.villageName,COUNT(s.ninjaId)
FROM village v
JOIN shinobi s on s.village=v.villageName
GROUP BY v.villageName
HAVING COUNT(s.ninjaId)>1

--39. Jutsu and their creators
SELECT s.NinjaID,j.jutsuName,j.jutsuType AS Jutsu_type,s.ninjaName AS Creator_name ,j.powerLevel
FROM shinobi s
JOIN created_by c ON s.ninjaId = c.NinjaID
JOIN jutsu j ON c.JutsuName=j.jutsuName;

--40.Clan and their leaders
SELECT l.NinjaId,c.clanName,s.ninjaName AS leader_name
FROM clan c
JOIN led_by l ON c.clanName=l.clanName
JOIN shinobi s on s.ninjaId=l.NinjaId;
      You, 9 hours ago • maybe now
--41.Ninjas from 2 specific villages
SELECT ninjaName,village
FROM shinobi
WHERE village = 'Leaf '
   OR village = 'Rain ';

--42.Villages having more ninjas than average
SELECT v.villageName,COUNT(s.ninjaId) AS NinjaCount
FROM village v
JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
HAVING COUNT(s.ninjaId) > (
    SELECT AVG(VillageCounts.Total)
```

	NinjaId	clanName	leader_name
1	13	Akimichi	Choji Akimichi
2	6	Hatake	Kakashi Hatake
3	25	Hoshigaki	Kisame Hoshigaki
4	26	Hyuga	Neji Hyuga
5	11	Nara	Shikamaru Nara
6	1	Sarutobi	Hiruzen Sarutobi
7	3	Senju	Tsunade Senju
8	8	Uchiha	Sasuke Uchiha
9	7	Uzumaki	Naruto Uzumaki
10	12	Yamanaka	Ino Yamanaka

41. Ninjas from 2 specific villages

```sql
--38. list of villages along with the total number of ninjas residing in each
SELECT v.villageName,COUNT(s.ninjaId)
FROM village v
JOIN shinobi s on s.village=v.villageName
GROUP BY v.villageName
HAVING COUNT(s.ninjaId)>1

--39. Jutsu and their creators
SELECT s.NinjaID,j.jutsuName,j.jutsuType AS Jutsu_type,s.ninjaName AS Creator_name ,j.powerLevel
FROM shinobi s
JOIN created_by c ON s.ninjaId = c.NinjaID
JOIN jutsu j ON c.JutsuName=j.jutsuName;

--40.Clan and their leaders
SELECT l.NinjaId,c.clanName,s.ninjaName AS leader_name
FROM clan c
JOIN led_by l ON c.clanName=l.clanName
JOIN shinobi s on s.ninjaId=l.NinjaId;

--41.Ninjas from 2 specific villages
SELECT ninjaName,village
FROM shinobi
WHERE village = 'Leaf '
   OR village = 'Rain ';
      You, 9 hours ago • maybe now
--42.Villages having more ninjas than average
SELECT v.villageName,COUNT(s.ninjaId) AS NinjaCount
FROM village v
JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
HAVING COUNT(s.ninjaId) > (
    SELECT AVG(VillageCounts.Total)
```

	ninjaName	village
1	Hiruzen Sarutobi	Leaf
2	Jiraiya	Leaf
3	Tsunade Senju	Leaf
4	Orochimaru	Leaf
5	Minato Namikaze	Leaf
6	Kakashi Hatake	Leaf
7	Naruto Uzumaki	Leaf
8	Sasuke Uchiha	Leaf
9	Sakura Haruno	Leaf
10	Asuma Sarutobi	Leaf
11	Shikamaru Nara	Leaf
12	Ino Yamanaka	Leaf
13	Choji Akimichi	Leaf
14	Might Guy	Leaf
15	Rock Lee	Leaf
16	Itachi Uchiha	Leaf
17	Pain	Rain
18	Konan	Rain
19	Tobi	Leaf
20	Neji Hyuga	Leaf
21	Hinata Hyuga	Leaf
22	Konohamaru Sarutobi	Leaf

42. Villages having more ninjas than average

Functionalities

```sql
--42.Villages having more ninjas than average
SELECT v.villageName,COUNT(s.ninjaId) AS NinjaCount
FROM village v
JOIN shinobi s ON v.villageName = s.village
GROUP BY v.villageName
HAVING COUNT(s.ninjaId) > (
    SELECT AVG(VillageCounts.Total)
    FROM (
        SELECT COUNT(ninjaId) AS Total
        FROM shinobi
        GROUP BY village
    ) AS VillageCounts
);
--43.Jutsus having more power than specific jutsu
SELECT jutsuName,jutsuType,powerLevel
FROM jutsu
WHERE powerLevel > ALL (
    SELECT powerLevel
    FROM jutsu
    WHERE jutsuType = 'Genjutsu'
);
--44.Squads with mentor names having more than 1 member under it
SELECT m.ninjaName AS MentorName, COUNT(s.ninjaId) AS Ninja_count
FROM shinobi m
JOIN shinobi s ON m.ninjaId = s.mentorNinjaId
GROUP BY m.ninjaName
HAVING COUNT(s.ninjaId) > 1;

--45.Village abandoned
SELECT villageName
FROM village
```

Results (1) Messages

	villageName	NinjaCount
1	Leaf	21

43. Jutsus having more power than specific jutsu

```sql
--43.Jutsus having more power than specific jutsu
SELECT jutsuName,jutsuType,powerLevel
FROM jutsu
WHERE powerLevel > ALL (
    SELECT powerLevel
    FROM jutsu
    WHERE jutsuType = 'Genjutsu'
);
--44.Squads with mentor names having more than 1 member under it
SELECT m.ninjaName AS MentorName, COUNT(s.ninjaId) AS Ninja_count
FROM shinobi m
JOIN shinobi s ON m.ninjaId = s.mentorNinjaId
GROUP BY m.ninjaName
HAVING COUNT(s.ninjaId) > 1;

--45.Village abandoned
SELECT villageName
FROM village
WHERE villageName NOT IN (
    SELECT villageName
    FROM presided_by
    WHERE LeaderShipStatus = 'Active'
);

--46.Ninja who isn't in standoff
SELECT ninjaName
FROM shinobi
WHERE ninjaId IN (SELECT ninjaId1 FROM standoff UNION SELECT ninjaId2 FROM standoff) -- Participated
AND ninjaId NOT IN (
    SELECT
        CASE
```

Results (1) Messages

	jutsuName	jutsuType	powerLevel
1	Almighty Push	Dojutsu	9.90
2	Amaterasu	Dojutsu	9.90
3	C4 Karura	Ninjutsu	9.10
4	Chidori	Ninjutsu	8.00
5	Daytime Tiger	Taijutsu	9.20
6	Fire Dragon Flame	Ninjutsu	7.50
7	Gentle Fisst	Taijutsu	7.50
8	Gentle Fist	Taijutsu	7.50
9	Heavenly Foot of Pain	Taijutsu	7.00
10	Hidden Lotus	Taijutsu	8.80
11	Human Bullet Tank	Taijutsu	6.80
12	Kamui	Dojutsu	9.70
13	Kirin	Ninjutsu	9.50
14	Lariat	Taijutsu	8.70
15	Mind Transfer	Hiden	6.00
16	Paper Shuriken	Ninjutsu	7.20
17	Rasengan	Ninjutsu	8.50
18	Sand Coffin	Ninjutsu	7.90
19	Sea Dragon	Ninjutsu	7.40
20	Shadow Stitching	Hiden	6.50
21	Snake Authority	Ninjutsu	8.20
22	Water Dragon	Ninjutsu	7.80

44. Squads with mentor names having more than 1 member under it

Functionalities

```sql
--43.Jutsus having more power than specific jutsu
SELECT jutsuName,jutsuType,powerLevel
FROM jutsu
WHERE powerLevel > ALL (
    SELECT powerLevel
    FROM jutsu
    WHERE jutsuType = 'Genjutsu'
);
--44.Squads with mentor names having more than 1 member under it
SELECT m.ninjaName AS MentorName, COUNT(s.ninjaId) AS Ninja_count
FROM shinobi m
JOIN shinobi s ON m.ninjaId = s.mentorNinjaId
GROUP BY m.ninjaName
HAVING COUNT(s.ninjaId) > 1;
--45.Village abandoned
SELECT villageName
FROM village
WHERE villageName NOT IN (
    SELECT villageName
    FROM presided_by
    WHERE LeaderShipStatus = 'Active'
);

--46.Ninja who isn't in standoff
SELECT ninjaName
FROM shinobi
WHERE ninjaId IN (SELECT ninjaId1 FROM standoff UNION SELECT ninjaId2 FROM standoff) -- Participated
AND ninjaId NOT IN (
    SELECT
        CASE
```

	MentorName	Ninja_count
1	Asuma Sarutobi	3
2	Hiruzen Sarutobi	5
3	Jiraiya	3
4	Kakashi Hatake	3
5	Might Guy	3
6	Minato Namikaze	2
7	Naruto Uzumaki	2

45. Village abandoned

```sql
--43.Jutsus having more power than specific jutsu
SELECT jutsuName,jutsuType,powerLevel
FROM jutsu
WHERE powerLevel > ALL (
    SELECT powerLevel
    FROM jutsu
    WHERE jutsuType = 'Genjutsu'
);
--44.Squads with mentor names having more than 1 member under it
SELECT m.ninjaName AS MentorName, COUNT(s.ninjaId) AS Ninja_count
FROM shinobi m
JOIN shinobi s ON m.ninjaId = s.mentorNinjaId
GROUP BY m.ninjaName
HAVING COUNT(s.ninjaId) > 1;

--45.Village abandoned
SELECT villageName
FROM village
WHERE villageName NOT IN (
    SELECT villageName
    FROM presided_by
    WHERE LeaderShipStatus = 'Active'
);
--46.Ninja who isn't in standoff
SELECT ninjaName
FROM shinobi
WHERE ninjaId IN (SELECT ninjaId1 FROM standoff UNION SELECT ninjaId2 FROM standoff) -- Participated
AND ninjaId NOT IN (
    SELECT
        CASE
```

	villageName
1	Cloud
2	Mist
3	Sound
4	Stone
5	Sugukure
6	Tagikakure

46. Ninja who isn't in standoff

Functionalities

```
--46.Ninja who isn't in standoff
SELECT ninjaName
FROM shinobi
WHERE ninjaId IN (SELECT ninjaId1 FROM standoff UNION SELECT ninjaId2 FROM standoff) ---Participated
AND ninjaId NOT IN (
    SELECT
        CASE
            WHEN ninjaId1 = winnerNinjaId THEN ninjaId2
            ELSE ninjaId1
        END
    FROM standoff
    WHERE winnerNinjaId IS NOT NULL
);
      You, 9 hours ago • maybe now
--47.Jutsu owned/used by no-one
SELECT j.jutsuName, j.jutsuType
FROM jutsu j
WHERE j.jutsuName NOT IN (
    SELECT DISTINCT s.jutsuName
    FROM shinobi s
    WHERE s.jutsuName IS NOT NULL
);

--48.Villages and their details
SELECT
    v.villageName,
    (SELECT COUNT(*) FROM shinobi s WHERE s.village = v.villageName) AS NinjaCount,
    (SELECT COUNT(*) FROM ninjaMission m WHERE m.clientNation = v.villageName) AS MissionCount,
    (SELECT COUNT(*) FROM rogueAssociation ra WHERE ra.originNationName = v.villageName) AS RogueAssocCount
FROM village v;
```

Results (1) Messages Open in New Tab

	ninjaName
1	Orochimaru
2	Minato Namikaze
3	Kakashi Hatake
4	Ino Yamanaka
5	Temari
6	Kankuro
7	Killer Bee
8	Pain

47. Jutsu owned/used by no-one

```
--46.Ninja who isn't in standoff
SELECT ninjaName
FROM shinobi
WHERE ninjaId IN (SELECT ninjaId1 FROM standoff UNION SELECT ninjaId2 FROM standoff) -- Participated
AND ninjaId NOT IN (
    SELECT
        CASE
            WHEN ninjaId1 = winnerNinjaId THEN ninjaId2
            ELSE ninjaId1
        END
    FROM standoff
    WHERE winnerNinjaId IS NOT NULL
);

--47.Jutsu owned/used by no-one
SELECT j.jutsuName, j.jutsuType
FROM jutsu j
WHERE j.jutsuName NOT IN (
    SELECT DISTINCT s.jutsuName
    FROM shinobi s
    WHERE s.jutsuName IS NOT NULL
);
      You, 10 hours ago • maybe now
--48.Villages and their details
SELECT
    v.villageName,
    (SELECT COUNT(*) FROM shinobi s WHERE s.village = v.villageName) AS NinjaCount,
    (SELECT COUNT(*) FROM ninjaMission m WHERE m.clientNation = v.villageName) AS MissionCount,
    (SELECT COUNT(*) FROM rogueAssociation ra WHERE ra.originNationName = v.villageName) AS RogueAssocCount
FROM village v;
```

Results (1) Messages Open in New Tab

	jutsuName	jutsuType
1	Gentle Fisst	Taijutsu

48. Villages and their details

Functionalities

```sql
--48.Villages and their details
SELECT
    v.villageName,
    (SELECT COUNT(*) FROM shinobi s WHERE s.village = v.villageName) AS NinjaCount,
    (SELECT COUNT(*) FROM ninjaMission m WHERE m.clientNation = v.villageName) AS MissionCount,
    (SELECT COUNT(*) FROM rogueAssociation ra WHERE ra.originNationName = v.villageName) AS RogueAssocCount
FROM village v;

--49.Jutsus and their details      You, 10 hours ago • maybe now
SELECT
    j.jutsuName,
    j.jutsuStatus,
    COALESCE(s_creator.ninjaName, 'Unknown') AS Creator,
    COUNT(s_users.ninjaId) AS NumberOfUsers
FROM jutsu j
LEFT JOIN created_by cb ON j.jutsuName = cb.JutsuName
LEFT JOIN shinobi s_creator ON cb.NinjaID = s_creator.ninjaId
LEFT JOIN shinobi s_users ON j.jutsuName = s_users.jutsuName
GROUP BY j.jutsuName, j.jutsuStatus, s_creator.ninjaName;

--50.Min cooldown time of jutsu
SELECT MIN(coolDownTime) AS Shortest_Cooldown
FROM jutsu;
SELECT TOP 5 jutsuName, coolDownTime
FROM jutsu
ORDER BY coolDownTime ASC;

--51.Clans having more than specific amount
SELECT clanName,clanSize,specialAbilities,originVillage
FROM clan
WHERE clanSize > 50
```

Results (1) Messages Open in New Tab

	villageName	NinjaCount	MissionCount	RogueAssocCount
1	Cloud	1	2	0
2	Leaf	21	6	1
3	Mist	1	2	1
4	Rain	2	1	2
5	Sand	3	2	0
6	Sound	0	0	0
7	Stone	1	1	0
8	Sugukure	0	0	0
9	Tagikakure	0	0	0

49. Jutsus and their details

```sql
--49.Jutsus and their details
SELECT
    j.jutsuName,
    j.jutsuStatus,
    COALESCE(s_creator.ninjaName, 'Unknown') AS Creator,
    COUNT(s_users.ninjaId) AS NumberOfUsers
FROM jutsu j
LEFT JOIN created_by cb ON j.jutsuName = cb.JutsuName
LEFT JOIN shinobi s_creator ON cb.NinjaID = s_creator.ninjaId
LEFT JOIN shinobi s_users ON j.jutsuName = s_users.jutsuName
GROUP BY j.jutsuName, j.jutsuStatus, s_creator.ninjaName;
      You, 10 hours ago • maybe now
--50.Min cooldown time of jutsu
SELECT MIN(coolDownTime) AS Shortest_Cooldown
FROM jutsu;
SELECT TOP 5 jutsuName, coolDownTime
FROM jutsu
ORDER BY coolDownTime ASC;

--51.Clans having more than specific amount
SELECT clanName,clanSize,specialAbilities,originVillage
FROM clan
WHERE clanSize > 50
ORDER BY clanSize DESC;

--52.Full clan details with missing details
SELECT c.clanName,c.specialAbilities,s.ninjaName AS LeaderName,lb.StartingDate
FROM clan c
FULL OUTER JOIN led_by lb ON c.clanName = lb.clanName
LEFT JOIN shinobi s ON lb.NinjaId = s.ninjaId;
```

Results (1) Messages Open in New Tab

	jutsuName	jutsuStatus	Creator	NumberOfUsers
1	Gentle Fisst	Clan Technique	Unknown	0
2	Human Bullet Tank	Clan Technique	Choji Akimichi	1
3	C4 Karura	Forbidden	Deidara	1
4	Sand Coffin	Active	Gaara	2
5	Fire Dragon Flame	Active	Hiruzen Sarutobi	2
6	Mind Transfer	Clan Technique	Ino Yamanaka	1
7	Amaterasu	Forbidden	Itachi Uchiha	1
8	Chidori	Active	Kakashi Hatake	1
9	Lariat	Active	Killer Bee	1
10	Water Dragon	Active	Kisame Hoshigaki	1
11	Paper Shuriken	Active	Konan	1
12	Daytime Tiger	Forbidden	Might Guy	1
13	Hidden Lotus	Forbidden	Might Guy	1
14	Rasengan	Active	Minato Namikaze	5
15	Gentle Fist	Clan Technique	Neji Hyuga	2
16	Snake Authority	Forbidden	Orochimaru	1
17	Almighty Push	Forbidden	Pain	1
18	Kirin	Active	Sasuke Uchiha	1
19	Shadow Stitching	Clan Technique	Shikamaru Nara	1
20	Sea Dragon	Active	Temari	1
21	Kamui	Forbidden	Tobi	1
22	Heavenly Foot of Pain	Active	Tsunade Senju	2

50. Min cooldown time of jutsu

Functionalities

```sql
LEFT JOIN created_by cb ON j.jutsuName = cb.JutsuName
LEFT JOIN shinobi s_creator ON cb.NinjaID = s_creator.ninjaId
LEFT JOIN shinobi s_users ON j.jutsuName = s_users.jutsuName
GROUP BY j.jutsuName, j.jutsuStatus, s_creator.ninjaName;

--50.Min cooldown time of jutsu
SELECT MIN(coolDownTime) AS Shortest_Cooldown      You, 10 hours ago • maybe now
FROM jutsu;

SELECT TOP 5 jutsuName, coolDownTime
FROM jutsu
ORDER BY coolDownTime ASC;

--51.Clans having more than specific amount
SELECT clanName,clanSize,specialAbilities,originVillage
FROM clan
WHERE clanSize > 50
ORDER BY clanSize DESC;

--52.Full clan details with missing details
SELECT c.clanName,c.specialAbilities,s.ninjaName AS LeaderName,lb.StartingDate
FROM clan c
FULL OUTER JOIN led_by lb ON c.clanName = lb.clanName
LEFT JOIN shinobi s ON lb.NinjaId = s.ninjaId;

--53.Missions sorted by revenue
SELECT missionId,missionObjective,missionRank,revenue
FROM ninjaMission
ORDER BY revenue DESC;

--54.Villages who have hosted chunin cometition and also have rogue associations originating from them
select v.* from village v
```

Results (2) Messages Open in New Tab

	Shortest_Cooldown
1	2.50

	jutsuName	coolDownTime
1	Human Bullet Tank	2.50
2	Paper Shuriken	2.80
3	Gentle Fisst	3.00
4	Gentle Fist	3.00
5	Heavenly Foot of Pain	3.00

51. Clans having more than specific amount

```sql
LEFT JOIN created_by cb ON j.jutsuName = cb.JutsuName
LEFT JOIN shinobi s_creator ON cb.NinjaID = s_creator.ninjaId
LEFT JOIN shinobi s_users ON j.jutsuName = s_users.jutsuName
GROUP BY j.jutsuName, j.jutsuStatus, s_creator.ninjaName;

--50.Min cooldown time of jutsu
SELECT MIN(coolDownTime) AS Shortest_Cooldown
FROM jutsu;

SELECT TOP 5 jutsuName, coolDownTime
FROM jutsu
ORDER BY coolDownTime ASC;

--51.Clans having more than specific amount
SELECT clanName,clanSize,specialAbilities,originVillage      You, 10 hours ago • maybe now
FROM clan
WHERE clanSize > 50
ORDER BY clanSize DESC;

--52.Full clan details with missing details
SELECT c.clanName,c.specialAbilities,s.ninjaName AS LeaderName,lb.StartingDate
FROM clan c
FULL OUTER JOIN led_by lb ON c.clanName = lb.clanName
LEFT JOIN shinobi s ON lb.NinjaId = s.ninjaId;

--53.Missions sorted by revenue
SELECT missionId,missionObjective,missionRank,revenue
FROM ninjaMission
ORDER BY revenue DESC;

--54.Villages who have hosted chunin cometition and also have rogue associations originating from them
select v.* from village v
```

Open in New Tab

	clanName	clanSize	specialAbilities	originVillage
1	Uchiha	150	Sharingan	Leaf
2	Uchihaa	150	Sharingan	Leaf
3	Hyuga	120	Byakugan	Leaf
4	Nara	80	Shadow Manipulation	Leaf
5	Akimichi	75	Body Expansion	Leaf
6	Yamanaka	70	Mind Transfer	Leaf
7	Sarutobi	60	Fire Release	Leaf
8	Senju	60	Immense Chakra and Wood Affinity	Leaf

52. Full clan details with missing details

Functionalities

```sql
LEFT JOIN created_by cb ON j.jutsuName = cb.JutsuName
LEFT JOIN shinobi s_creator ON cb.NinjaID = s_creator.ninjaId
LEFT JOIN shinobi s_users ON j.jutsuName = s_users.jutsuName
GROUP BY j.jutsuName, j.jutsuStatus, s_creator.ninjaName;

--50.Min cooldown time of jutsu
SELECT MIN(coolDownTime) AS Shortest_Cooldown
FROM jutsu;

SELECT TOP 5 jutsuName, coolDownTime
FROM jutsu
ORDER BY coolDownTime ASC;

--51.Clans having more than specific amount
SELECT clanName,clanSize,specialAbilities,originVillage
FROM clan
WHERE clanSize > 50
ORDER BY clanSize DESC;

--52.Full clan details with missing details
SELECT c.clanName,c.specialAbilities,s.ninjaName AS LeaderName,lb.StartingDate
FROM clan c
FULL OUTER JOIN led_by lb ON c.clanName = lb.clanName
LEFT JOIN shinobi s ON lb.NinjaId = s.ninjaId;

--53.Missions sorted by revenue
SELECT missionId,missionObjective,missionRank,revenue
FROM ninjaMission
ORDER BY revenue DESC;

--54.Villages who have hosted chunin cometition and also have rogue associations originating from them
select v.* from village v
```

	clanName	specialAbilities	LeaderName	StartingDate
1	Akimichi	Body Expansion	Choji Akimichi	1999-05-01
2	Amegakure Elite	Rain Sensory Techniques	NULL	NULL
3	Explosion Corps	Explosion Release	NULL	NULL
4	Hatake	Elite Combat Skill and Tactical Intelligence	Kakashi Hatake	1972-09-15
5	Hoshigaki	Enhanced Strength and Chakra	Kisame Hoshigaki	1980-03-18
6	Hyuga	Byakugan	Neji Hyuga	1999-07-03
7	Kazekage Line	Sand Manipulation	NULL	NULL
8	Nara	Shadow Manipulation	Shikamaru Nara	1999-09-22
9	Otogakure Experiments	Curse Mark Enhancement	NULL	NULL
10	Sarutobi	Fire Release	Hiruzen Sarutobi	1937-02-08
11	Senju	Immense Chakra and Wood Affinity	Tsunade Senju	1950-08-02
12	Uchiha	Sharingan	Sasuke Uchiha	1999-07-23
13	Uchihaa	Sharingan	NULL	NULL
14	Uzumaki	Massive Chakra Reserves and Sealing Jutsu	Naruto Uzumaki	1999-10-10
15	Yamanaka	Mind Transfer	Ino Yamanaka	1999-09-23
16	Yotsuki	Lightning Release Mastery	NULL	NULL

53. Missions sorted by revenue

```sql
LEFT JOIN created_by cb ON j.jutsuName = cb.JutsuName
LEFT JOIN shinobi s_creator ON cb.NinjaID = s_creator.ninjaId
LEFT JOIN shinobi s_users ON j.jutsuName = s_users.jutsuName
GROUP BY j.jutsuName, j.jutsuStatus, s_creator.ninjaName;

--50.Min cooldown time of jutsu
SELECT MIN(coolDownTime) AS Shortest_Cooldown
FROM jutsu;

SELECT TOP 5 jutsuName, coolDownTime
FROM jutsu
ORDER BY coolDownTime ASC;

--51.Clans having more than specific amount
SELECT clanName,clanSize,specialAbilities,originVillage
FROM clan
WHERE clanSize > 50
ORDER BY clanSize DESC;

--52.Full clan details with missing details
SELECT c.clanName,c.specialAbilities,s.ninjaName AS LeaderName,lb.StartingDate
FROM clan c
FULL OUTER JOIN led_by lb ON c.clanName = lb.clanName
LEFT JOIN shinobi s ON lb.NinjaId = s.ninjaId;

--53.Missions sorted by revenue
SELECT missionId,missionObjective,missionRank,revenue
FROM ninjaMission
ORDER BY revenue DESC;

--54.Villages who have hosted chunin cometition and also have rogue associations originating from them
select v.* from village v
```

	missionId	missionObjective	missionRank	revenue
1	103	Infiltrate Hidden Rain	S	200000.00
2	108	Intercept Akatsuki Spy	S	180000.00
3	102	Retrieve Forbidden Scroll	S	150000.00
4	114	Sabotage Weapon Supply Line	A	90000.00
5	106	Protect Land of Fire Daimyo	A	80000.00
6	113	Reconnaissance in Land of Iron	A	70000.00
7	101	Escort Bridge Builder Tazuna	A	50000.00
8	107	Investigate Rogue Activity	B	45000.00
9	104	Defend Sunagakure Borders	B	35000.00
10	109	Deliver Secret Intelligence	B	30000.00
11	112	Neutralize Bandit Leader	B	25000.00
12	110	Search for Rare Ore	C	15000.00
13	105	Capture Rare Medicinal Herbs	C	12000.00
14	111	Escort Merchant Caravan	C	10000.00
15	115	Recover Stolen Kunai Shipment	D	5000.00

54. Villages who have hosted chunin competition and also have rogue associations originating from them

Functionalities

```
--54.Villages who have hosted chunin cometition and also have rogue associations originating from them
select v.* from village v
join chuninCompetition c
on v.villageName = c.venueNation

INTERSECT

select v.* from village v
join rogueAssociation r
on v.villageName = r.originNationName

--55. Display ninjas who never won a standoff but participated in atleast one
select n.* from shinobi n
join standoff s
on s.ninjaId1 = n.ninjaId
or s.ninjaId2 = n.ninjaId
where not exists
        (
            select 1
            from standoff d
            where n.ninjaId = d.winnerNinjaId
        )

--56. clans who never sponsored chunin competitions
select * from clan
except
select c.* from clan c
join chuninCompetition cc
on c.clanName = cc.MainSponsorClan

--57. all items owned by nations and region including those owned by none
```

Results (1) Messages

	villageName	region	foundedYear	VillagePopulation	tailBeast	powerInd
1	Leaf	Land of Fire	1937	1230	9-Tails	9.50

55. Display ninjas who never won a standoff but participated in atleast one

```
--55. Display ninjas who never won a standoff but participated in atleast one
select n.* from shinobi n
join standoff s
on s.ninjaId1 = n.ninjaId
or s.ninjaId2 = n.ninjaId
where not exists
        (
            select 1
            from standoff d
            where n.ninjaId = d.winnerNinjaId
        )

--56. clans who never sponsored chunin competitions
select * from clan
except
select c.* from clan c
join chuninCompetition cc
on c.clanName = cc.MainSponsorClan

--57. all items owned by nations and region including those owned by none
select i.*,v.region from village v
right join inventory i
on i.ownedByNation = v.villageName

--58. Find all jutsu that contain word "fist" in them
select * from jutsu
where jutsuName like '%fist%'
```

Results (1) Messages

	ninjaId	ninjaName	dateOfBirth	NinjaStatus	clan	chakraNature
1	1	Hiruzen Sarutobi	1937-02-08	Deceased	Sarutobi	Fire
2	1	Hiruzen Sarutobi	1937-02-08	Deceased	Sarutobi	Fire
3	2	Jiraiya	1946-11-11	Deceased	NULL	Fire
4	9	Sakura Haruno	1999-03-28	Alive	NULL	Earth
5	9	Sakura Haruno	1999-03-28	Alive	NULL	Earth
6	10	Asuma Sarutobi	1970-10-18	Deceased	Sarutobi	Wind
7	12	Ino Yamanaka	1999-09-23	Alive	Yamanaka	Yin
8	14	Might Guy	1970-01-01	Alive	NULL	Fire
9	15	Rock Lee	2000-11-27	Alive	NULL	None
10	20	Itachi Uchiha	1985-06-09	Deceased	Uchiha	Fire
11	23	Tobi	1983-02-10	Deceased	Uchiha	None
12	23	Tobi	1983-02-10	Deceased	Uchiha	None
13	24	Deidara	1988-05-05	Deceased	NULL	Earth
14	25	Kisame Hoshigaki	1980-03-18	Deceased	Hoshigaki	Water
15	25	Kisame Hoshigaki	1980-03-18	Deceased	Hoshigaki	Water
16	27	Hinata Hyuga	1999-12-27	Alive	Hyuga	Fire

56. Clans who never sponsored chunin competitions

Functionalities

```
--56. clans who never sponsored chunin competitions
select * from clan
except
select c.* from clan c
join chuninCompetition cc
on c.clanName = cc.MainSponsorClan

--57. all items owned by nations and region including those owned by none
select i.*,v.region from village v
right join inventory i
on i.ownedByNation = v.villageName

--58. Find all jutsu that contain word "fist" in them
select * from jutsu
where jutsuName like '%fist%'
```

Results (1) Messages

	clanName	originVillage	specialAbilities	clanSymbol	clanSize
1	Akimichi	Leaf	Body Expansion	Butterfly	75
2	Amegakure Elite	Rain	Rain Sensory Techniques	Rain Drop	30
3	Explosion Corps	Stone	Explosion Release	Blast Mark	45
4	Hatake	Leaf	Elite Combat Skill and Tactical Intelligence	White Fang Symbol	15
5	Hoshigaki	Mist	Enhanced Strength and Chakra	Shark	35
6	Kazekage Line	Sand	Sand Manipulation	Gourd	40
7	Otogakure Experiments	Sound	Curse Mark Enhancement	Sound Wave	25
8	Sarutobi	Leaf	Fire Release	Monkey	60
9	Senju	Leaf	Immense Chakra and Wood Affinity	Leaf Crest	60
10	Uchihaa	Leaf	Sharingan	Fan	150
11	Yamanaka	Leaf	Mind Transfer	Flower	70
12	Yotsuki	Cloud	Lightning Release Mastery	Lightning Crest	50

57. All items owned by nations and region including those owned by none

```
--56. clans who never sponsored chunin competitions
select * from clan
except
select c.* from clan c
join chuninCompetition cc
on c.clanName = cc.MainSponsorClan

--57. all items owned by nations and region including those owned by none
select i.*,v.region from village v
right join inventory i
on i.ownedByNation = v.villageName

--58. Find all jutsu that contain word "fist" in them
select * from jutsu
where jutsuName like '%fist%'
```

	ninjaGearId	Iname	Irange	worth	Icount	rarity	owne
1	1	Standard Kunai	5.50	0.50	5000	Common	Leaf
2	2	Paper Bomb	3.00	1.20	1200	Common	Leaf
3	3	Smoke Bomb	2.00	0.80	850	Medium	Sand
4	4	Shuriken	7.50	0.30	8000	Common	Cloud
5	5	Giant Fan	9.99	5.50	15	Rare	Sand
6	6	Samehada Scale	1.50	9.00	1	Legendary	Mist
7	7	Hidan Blade	1.00	2.50	300	Rare	Stone
8	8	Poison Senbon	6.00	1.50	2000	Rare	Sand
9	9	Flash Bomb	4.00	1.10	600	Common	Cloud
10	10	Chakra Blade	2.00	8.50	50	Medium	Leaf
11	11	Zabuza Executioner Blade	2.50	8.50	1	Legendary	NULL
12	111	Zabuza Executioner Blade	2.50	8.50	1	Legendary	NULL

58. Find all jutsu that contain word "fist" in them

Functionalities

```
--56. clans who never sponsored chunin competitions
select * from clan
except
select c.* from clan c
join chuninCompetition cc
on c.clanName = cc.MainSponsorClan

--57. all items owned by nations and region including those owned by none
select i.*,v.region from village v
right join inventory i
on i.ownedByNation = v.villageName

--58. Find all jutsu that contain word "fist" in them
select * from jutsu        You, 10 hours ago • maybe now
where jutsuName like '%fist%'
```

Open in New Tab

	jutsuName	jutsuType	coolDownTime	powerLevel	jutsuStatus	chakraCo
1	Gentle Fist	Taijutsu	3.00	7.50	Clan Technique	Medium